

components to make a final product. As companies scale up to make components, they will rely more on automated processes and robots,” says J.S. Gujral, managing director, Syrma SGS Technology, a Gurugram-based electronics manufacturing company which is setting up India’s largest Printed Circuit Board (PCB) plant at Tirupati, Andhra Pradesh.

To be globally competitive and scale up, companies will need newer, more automated, factories.

For example, beyond the main processor in a phone, there are hundreds of components mounted on PCBs. To make these precision components, factories need to upgrade with robots that can multi-task. A line with robots might have no humans and could essentially run in the ‘dark mode’; such facilities are called dark factories, as no people working inside would mean that the factory may not need lights.

74 |

“There could be some version of hybrid dark factories now, but as component manufacturing increases, more robots and automated processes will be used,” says Gujral.

## NEW VISION

The image of factories dominated by machines, metal and men could soon be replaced with more robots and automated processes. At present, some of the tasks in factories have been automated. There are a few robots here and there, but

much of the heavy lifting is still labour-intensive.

Cut to the modern factory, and that image is fast disappearing. Even before a single brick is laid or a machine ordered, companies are creating virtual twins to see what will be placed where, conduct stress tests, assess sustainability, and so on.

And these are just a few ways in which factories of tomorrow,

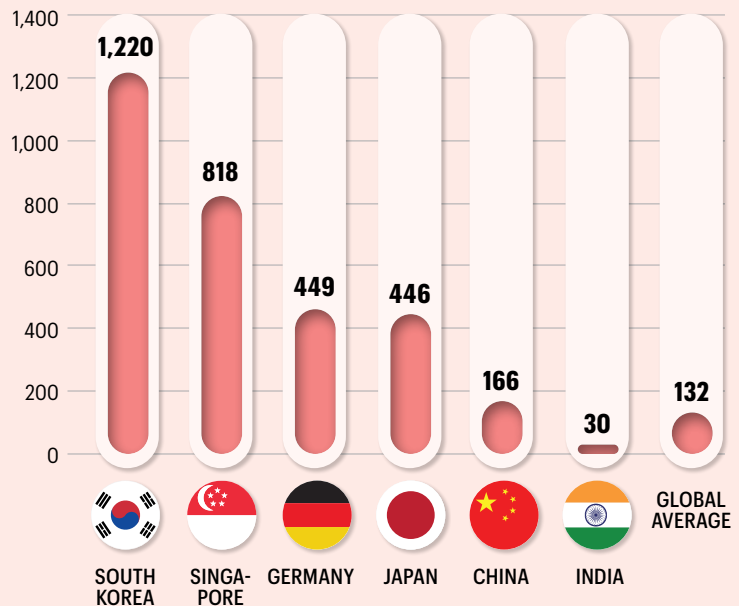
churning out everything from handsets to smart speakers, are being built. The good part is that India is beginning to manufacture at a global scale, and units here can now leapfrog with some of the latest innovations to build future-ready factories.

Tejpreet Chopra, founder & CEO, industry.ai (which creates AI solutions for manufacturing), says, “We don’t have a choice but

# ROBOTS IN MANUFACTURING INDUSTRIES

• South Korea is a world leader in automation. Singapore has high robot density because of small population

## ROBOT DENSITY (UNITS PER 10,000 WORKERS)



**NOTE** ROBOT DENSITY IS THE NUMBER OF OPERATIONAL ROBOTS RELATIVE TO THE NUMBER OF EMPLOYEES **SOURCE** INTERNATIONAL FEDERATION OF ROBOTICS, 2025

## FACTORIES GET A MAKEOVER

**+** More robots, enhanced by AI, are being deployed on factory floors

**+** Tasks like visual inspection are being replaced by computer vision

**+** Robots and humans work together in hybrid plants, eventually leading to ‘dark factories’